

Electricity and Magnetism

Our understanding of electric current must begin with the nature of matter. All matter is composed of molecules. All molecules are made up of atoms, which are themselves made up of electrons, protons, and neutrons.

Electricity is the presence and motion of charged particles. Electricity is often described as being either static or dynamic.

The difference between the two is based simply on whether the electrons are at rest (static) or in motion (dynamic).

Static electricity is a build up of an electrical charge on the surface of an object. It is considered "static" due to the fact that there is no current flowing as in AC or DC electricity. Static electricity is usually caused when non-conductive materials such as rubber, plastic or glass are rubbed together, causing a transfer of electrons, which then results in an imbalance of charges between the two materials. The fact that there is an imbalance of charges between the two materials means that the objects will exhibit an attractive or repulsive force.

History

The physics of electromagnetism was first studied by the early Greek philosophers, who discovered that if a piece of amber is rubbed and then brought near bits of straw, the straw will jump to the amber. We now know that the attraction between amber and straw is due to an electric force. The Greek philosophers also discovered that if a certain type of stone (a naturally occurring magnet) is brought near bits of iron, the iron will jump to the stone. We now know that the attraction between magnet and iron is due to a magnetic force.

From these modest origins with the Greek philosophers, the sciences of electricity and magnetism developed separately for centuries—until 1820, in fact, when Hans Christian Oersted found a connection between them: an electric current in a wire can deflect a magnetic compass needle.

The new science of electromagnetism was developed further by workers in many countries. One of the best was Michael Faraday. His name is associated with electromagnetic induction. He discovered that a moving magnetic field induces electric current in a neighboring wire.

In the mid-nineteenth century, James Clerk Maxwell put Faraday's ideas into mathematical form, introduced many new ideas of his own.

Electric Charge Definition

Electric charge can be defined as a fundamental property of subatomic particles that gives rise to the phenomenon of experiencing force in the presence of electric and magnetic fields.

The vast amount of charge in an everyday object is usually hidden because the object contains equal amounts of the two kinds of charge: positive charge and negative charge. With such an equality—or balance—of charge, the object is said to be **electrically neutral**; that is, it contains no net charge

In atoms, the electron carries a negative elementary or unit charge; the proton carries a positive charge. The two types of charge are equal and opposite. In an atom of matter, an electrical charge occurs whenever the number of protons in the nucleus differs from the number of electrons surrounding that nucleus. If there are more electrons than protons, the atom has a negative charge.

Charged objects interact by exerting forces on one another.

we first charge a glass rod by rubbing one end with silk.

Suppose we now suspend the charged rod from a thread to electrically isolate it from its surroundings so that its charge cannot change

If we bring a second, similarly charged, glass rod nearby the two rods repel each other; that is, each rod experiences a force directed away from the other rod

if we rub a plastic rod with fur and then bring the rod near the suspended glass rod the two rods attract each other

Later on we shall put this rule into quantitative form as Coulomb's law of electrostatic force (or electric force) between charges.

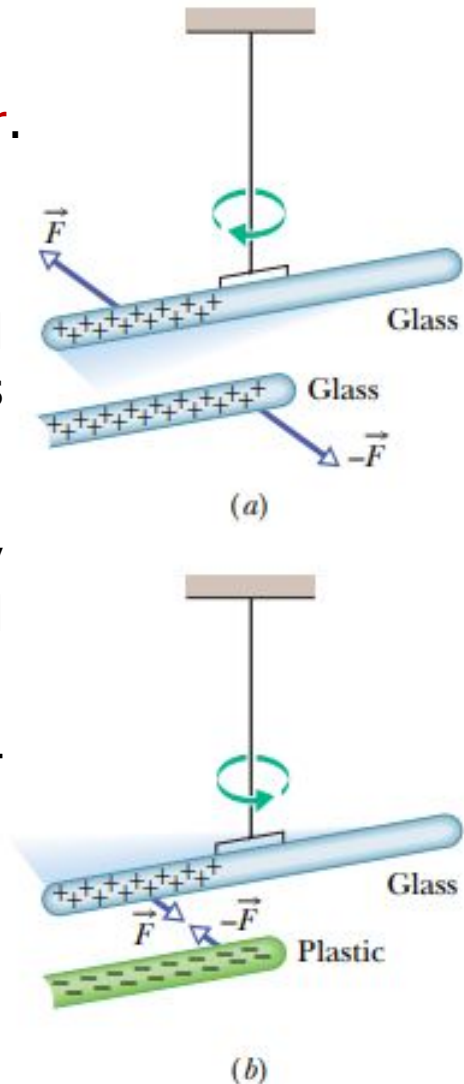


Fig. 21-2 (a) Two charged rods of the same sign repel each other. (b) Two charged rods of opposite signs attract each other. Plus signs indicate a positive net charge, and minus signs indicate a negative net charge.

Conductors and Insulators

Based on the ability of charge to move through them we can classify materials as follows:

Conductors are materials through which charge can move rather freely; examples include metals (such as copper in common lamp wire), the human body, and tap water.

Nonconductors—also called insulators—are materials through which charge cannot move freely; examples include rubber (such as the insulation on common lamp wire), plastic, glass, and chemically pure water.

Semiconductors are materials that are intermediate between conductors and insulators; examples include silicon and germanium in computer chips.

The properties of conductors and insulators are due to the structure and electrical nature of atoms.

Atoms consist of positively charged protons, negatively charged electrons, and electrically neutral neutrons. The protons and neutrons are packed tightly together in a central nucleus.

The charge of a single electron and that of a single proton have the same magnitude but are opposite in sign. Hence, **an electrically neutral atom contains equal numbers of electrons and protons**. Electrons are held near the nucleus because they have the electrical sign opposite that of the protons in the nucleus and thus are attracted to the nucleus.

When atoms of a conductor like copper come together to form the solid, some of their outermost (and so most loosely held) electrons become free to wander about within the solid, leaving behind positively charged atoms (positive ions). We call the mobile electrons conduction electrons.

Here is an example of how conduction can eliminate excess charge on an object.

- If you rub a copper rod with wool, charge is transferred from the wool to the rod.
- However, if you are holding the rod while also touching a faucet, you cannot charge the rod in spite of the transfer.
- The reason is that you, the rod, and the faucet are all conductors connected, via the plumbing, to Earth's surface, which is a huge conductor.
- Because the excess charges put on the rod by the wool repel one another, they move away from one another by moving first through the rod, then through you, and then through the faucet and plumbing to reach Earth's surface, where they can spread out. The process leaves the rod electrically neutral.
- In thus setting up a pathway of conductors between an object and Earth's surface, we are said to ground the object, and in neutralizing the object (by eliminating an unbalanced positive or negative charge), we are said to discharge the object.
- If instead of holding the copper rod in your hand, you hold it by an insulating handle, you eliminate the conducting path to Earth, and the rod can then be charged by rubbing.

Coulomb's law

If two charged particles are brought near each other, they each exert a force on the other.

If the particles have the same sign of charge, they repel each other (Figs. 21-6a and b). That is, the force on each particle is directed away from the other particle, and if the particles can move, they move away from each other.

If, instead, the particles have opposite signs of charge, they attract each other (Fig. 21-6c) and, if free to move, they move closer to each other.

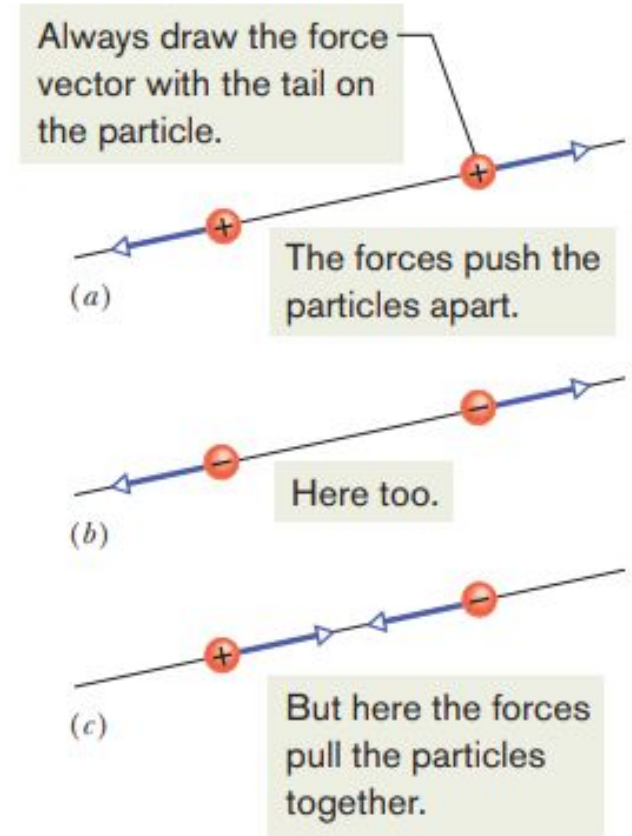


Fig. 21-6 Two charged particles repel each other if they have the same sign of charge, either (a) both positive or (b) both negative. (c) They attract each other if they have opposite signs of charge.

This force of repulsion or attraction due to the charge properties of objects is called an **electrostatic force**. The equation giving the force for charged particles is called Coulomb's law after Charles-Augustin de Coulomb, whose experiments in 1785 led him to it. In terms of the particles in Fig. 21-7, where particle 1 has charge q_1 and particle 2 has charge q_2 , the force on particle 1 is (Coulomb's law)

$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r} \quad \text{..... 1}$$

in which $\hat{r}$ is a unit vector along an axis extending through the two particles, r is the distance between them, and k is a constant.

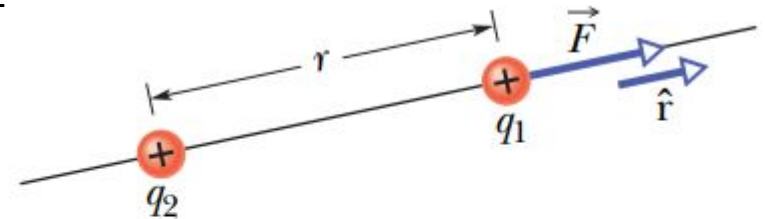


Fig. 21-7 The electrostatic force on particle 1 can be described in terms of a unit vector $\hat{r}$ along an axis through the two particles.

Above equation is the same as that of Newton's equation for the gravitational force between two particles with masses m_1 and m_2 that are separated by a distance r :

$$\vec{F} = G \frac{m_1 m_2}{r^2} \hat{r} \quad \text{..... 2} \quad \text{in which } G \text{ is the gravitational constant}$$

Both equations describe **inverse square laws** that involve a property of the interacting particles—the mass in one case and the charge in the other. The laws differ in that **gravitational forces are always attractive but electrostatic forces may be either attractive or repulsive**, depending on the signs of the two charges. This difference arises from the fact that, although there is only one kind of mass, there are two kinds of charge.

This simple law correctly accounts for the forces that bind atoms together to form molecules, and for the forces that bind atoms and molecules together to form solids and liquids.

The magnitude of force in Coulomb's law becomes

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2} \quad (\text{Coulomb's law}). \quad \text{..... 3}$$

The constants in Eqs. 1 and 3 have the value

$$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2. \quad \text{..... 4}$$

The quantity ϵ_0 , called the **permittivity constant**, sometimes appears separately in equations and is

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2. \quad \text{..... 5}$$

Still another parallel between the gravitational force and the electrostatic force is that both obey the principle of superposition. If we have n charged particles, they interact independently in pairs, and the force on any one of them, let us say particle 1, is given by the vector sum

$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \vec{F}_{15} + \cdots + \vec{F}_{1n}, \quad \text{..... 6}$$

in which, for example, $\vec{F}_{14}$ is the force acting on particle 1 due to the presence of particle 4. An identical formula holds for the gravitational force.

Unit of charge

The SI unit of charge is the coulomb. For practical reasons having to do with the accuracy of measurements, the coulomb unit is derived from the SI unit ampere for electric current i . Current is the rate dq/dt at which charge moves past a point or through a region.

$$i = dq/dt$$

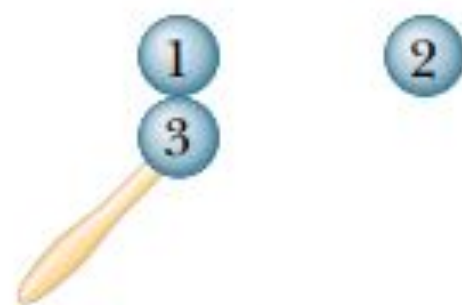
in which i is the current (in amperes) and dq (in coulombs) is the amount of charge moving past a point or through a region in time dt (in seconds). Rearranging Eq. tells us that

$$1 \text{ C} = (1 \text{ A})(1 \text{ s})$$

•2 ³⁸ Identical isolated conducting spheres 1 and 2 have equal charges and are separated by a distance that is large compared with their diameters (Fig. 21-21*a*). The electrostatic force acting on sphere 2 due to sphere 1 is $\vec{F}$. Suppose now that a third identical sphere 3, having an insulating handle and initially neutral, is touched first to sphere 1 (Fig. 21-21*b*), then to sphere 2 (Fig. 21-21*c*), and finally removed (Fig. 21-21*d*). The electrostatic force that now acts on sphere 2 has magnitude F' . What is the ratio F'/F ?



(a)



(b)



- 3 What must be the distance between point charge $q_1 = 26.0 \mu\text{C}$ and point charge $q_2 = -47.0 \mu\text{C}$ for the electrostatic force between them to have a magnitude of 5.70 N ?
- 4 In the return stroke of a typical lightning bolt, a current of $2.5 \times 10^4 \text{ A}$ exists for $20 \mu\text{s}$. How much charge is transferred in this event?
- 6 Two equally charged particles are held $3.2 \times 10^{-3} \text{ m}$ apart and then released from rest. The initial acceleration of the first particle is observed to be 7.0 m/s^2 and that of the second to be 9.0 m/s^2 . If the mass of the first particle is $6.3 \times 10^{-7} \text{ kg}$, what are (a) the mass of the second particle and (b) the magnitude of the charge of each particle?
- 14 Three particles are fixed on an x axis. Particle 1 of charge q_1 is at $x = -a$, and particle 2 of charge q_2 is at $x = +a$. If their net electrostatic force on particle 3 of charge $+Q$ is to be zero, what must be the ratio q_1/q_2 when particle 3 is at (a) $x = +0.500a$ and (b) $x = +1.50a$?

Solution of problem 14

Case a

$$\frac{1}{4\pi\epsilon_0} \frac{|q_1|Q}{(-a-\frac{a}{2})^2} = \frac{1}{4\pi\epsilon_0} \frac{|q_2|Q}{(a-\frac{a}{2})^2}$$

leads to $|q_1| = 9.0 |q_2|$. Q is located between q_1 and q_2 , we conclude q_1 and q_2 are like-sign. Consequently, $q_1/q_2 = 9.0$.

Case b

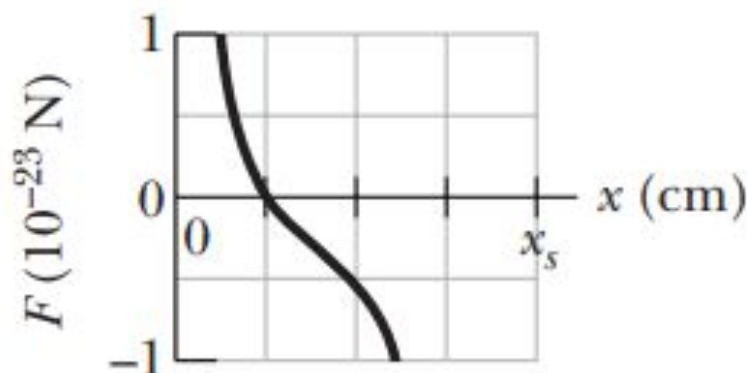
$$\frac{1}{4\pi\epsilon_0} \frac{|q_1|Q}{(-a-\frac{3a}{2})^2} = \frac{1}{4\pi\epsilon_0} \frac{|q_2|Q}{(a-\frac{3a}{2})^2}$$

yields $|q_1| = 25.0 |q_2|$. Now, Q is not located between q_1 and q_2 ; one of them must push and the other must pull. Thus, they are unlike-sign, so $q_1/q_2 = -25$.

••16 In Fig. 21-26a, particle 1 (of charge q_1) and particle 2 (of charge q_2) are fixed in place on an x axis, .8 cm apart. Particle 3 (of charge $q_3 = +8.00 \times 10^{-19}$ C) is to be placed on the line between particles 1 and 2 so that they produce a net electrostatic force $\vec{F}_{3,\text{net}}$ on it. Figure 21-26b gives the x component of that force versus the coordinate x at which particle 3 is placed. The scale of the x axis is set by $x_s = 8.0$ cm. What are (a) the sign of charge q_1 and (b) the ratio q_2/q_1 ?



(a)



(b)

Electric charge is quantized: any charge can be written as $q = ne$, where n is a positive or negative integer and e is a constant of nature called the elementary charge)

•24 Two tiny, spherical water drops, with identical charges of $-1.00 \times 10^{-16} \text{ C}$, have a center-to-center separation of 1.00 cm. (a) What is the magnitude of the electrostatic force acting between them? (b) How many excess electrons are on each drop, giving it its charge imbalance?

•27 The magnitude of the electrostatic force between two identical ions that are separated by a distance of $5.0 \times 10^{-10} \text{ m}$ is $3.7 \times 10^{-9} \text{ N}$. (a) What is the charge of each ion? (b) How many electrons are “missing” from each ion (thus giving the ion its charge imbalance)?

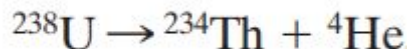
•28 _____ A current of 0.300 A through your chest can send your heart into fibrillation, ruining the normal rhythm of heartbeat and disrupting the flow of blood (and thus oxygen) to your brain. If that current persists for 2.00 min , how many conduction electrons pass through your chest?

Charge is conserved

If you rub a glass rod with silk, a positive charge appears on the rod. Measurement shows that a negative charge of equal magnitude appears on the silk. This suggests that rubbing does not create charge but only transfers it from one body to another, upsetting the electrical neutrality of each body during the process.

Thus, we add electric charge to our list of quantities—including energy and both linear and angular momentum—that obey a conservation law.

Important examples of the conservation of charge occur in the radioactive decay of nuclei, in which a nucleus transforms into (becomes) a different type of nucleus



The parent nucleus ^{238}U contains 92 protons (a charge of $92e$), the daughter nucleus ^{234}Th contains 90 protons (a charge of $90e$), and the emitted alpha particle ^4He contains 2 protons (a charge of $2e$). We see that the total charge is $92e$ before and after the decay; thus, charge is conserved. (The total number of protons and neutrons is also conserved: 238 before the decay and $234 + 4 = 238$ after the decay.)

41 (a) What equal positive charges would have to be placed on Earth and on the Moon to neutralize their gravitational attraction?
(b) Why don't you need to know the lunar distance to solve this problem?

44 How far apart must two protons be if the magnitude of the electrostatic force acting on either one due to the other is equal to the magnitude of the gravitational force on a proton at Earth's surface?

59. What is the total charge in coulombs of 75.0 kg of electrons?